

Proteograph

Typst proteomics package

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ABSTRACT

proteograph is a package defining [Typst](#) functions dedicated to proteomics.

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REFERENCE

XIC (eXtracted Ion Chromatogram) plot

Example :

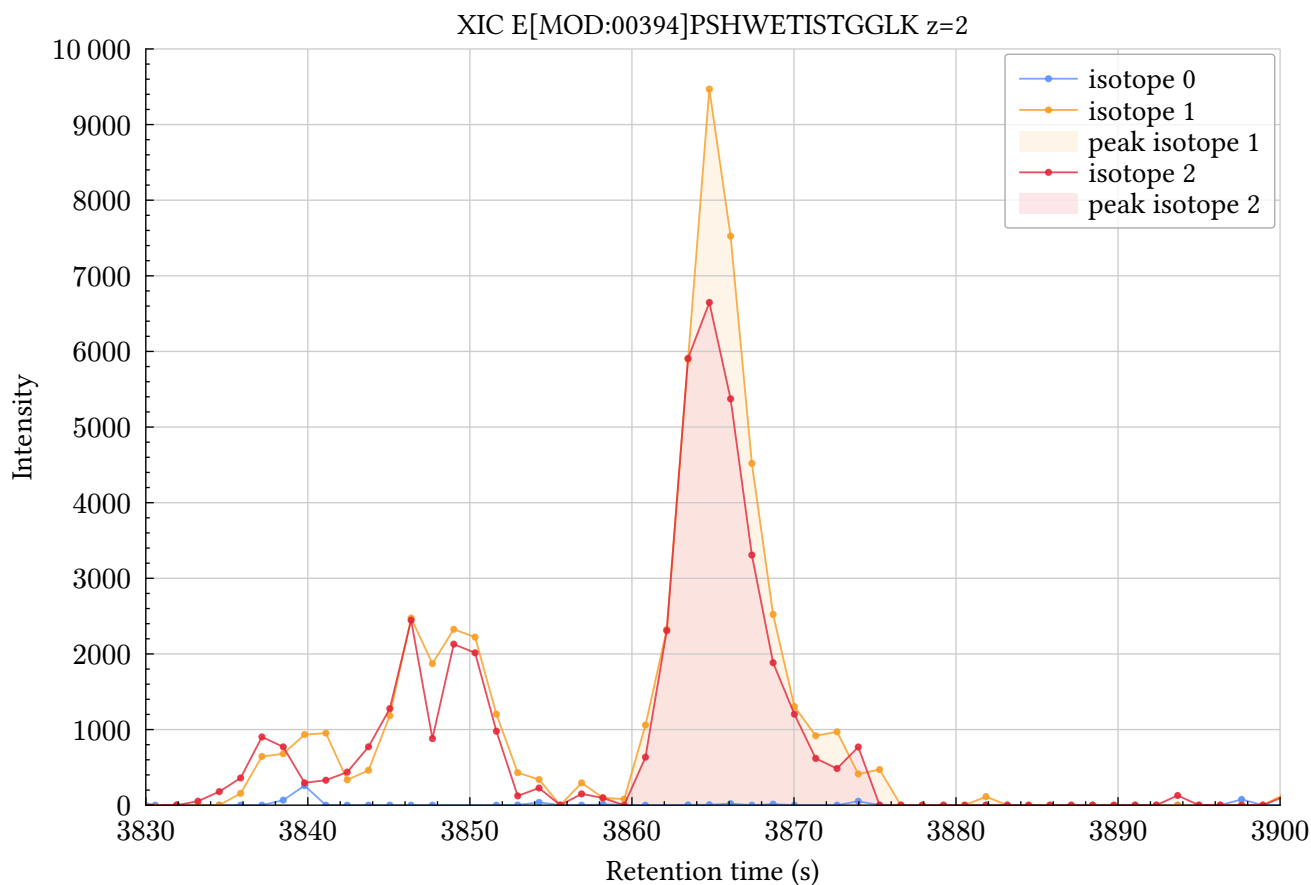
```
#import "@preview/tegraph:0.2.2": *

#let data_json = json("../examples/data/pepalb35.json")

#let xic0 = data_json.found_list_first_pass.first().xics.first().trace
#xic0.insert("title","isotope 0")
#let xic1 = data_json.found_list_first_pass.first().xics.at(1).trace
#xic1.insert("title","isotope 1")
#xic1.insert("peak-
begin",data_json.found_list_first_pass.first().xics.at(1).peak.rt.first())
#xic1.insert("peak-
end",data_json.found_list_first_pass.first().xics.at(1).peak.rt.last())
#let xic2 = data_json.found_list_first_pass.first().xics.at(2).trace
#xic2.insert("title","isotope 2")
#xic2.insert("peak-
begin",data_json.found_list_first_pass.first().xics.at(2).peak.rt.first())
#xic2.insert("peak-
end",data_json.found_list_first_pass.first().xics.at(2).peak.rt.last())

#xic-plot(height: 10cm, title: "XIC E[MOD:00394]PSHWETISTGGLK z=2", rt-range: (3830,
3900), max-intensity: 10000, xic0,xic1, xic2)
```

Result :



XIC plot full documentation

- [xic-isotope-plot\(\)](#)
- [xic-plot\(\)](#)

xic-isotope-plot

Generates a XIC plot separating isotope pattern by charge

Parameters

```
xic-isotope-plot(  
    width: length relative ,  
    height: length relative ,  
    rt-range: auto array ,  
    max-intensity: none float ,  
    title: content ,  
    ..xic-item: dictionary  
) -> content
```

width length or relative

The width of the diagram. This can be

- A `length` ; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto` .

Default: `15cm`

height length or relative

The height of the diagram. This can be

- A `length` ; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto` .

Default: `10cm`

rt-range auto or array

Data limits along the x-axis (retention time in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

max-intensity `none` or `float`

Maximum intensity to display. **Optional**.

Example: `30000`

Default: `none`

title `content`

Graph title

Default: `none`

..xic-item `dictionary`

dictionary containing XIC data

```
1  /// XIC dictionary structure typc
2  #let xic_item = (
3    /// Array of retention times values
4    "x": (),
5    /// Array of intensity values
6    "y": (),
7    /// Title for this XIC
8    "title": "title for this XIC"
9    /// Detected peak retention time start. *Optional*.
10   "peak-begin": 3500,
11   /// Detected peak retention time end. *Optional*.
12   "peak-end": 3500,
13   /// Charge.
14   "charge": none,
15   /// Isotope number.
16   "isotope": none,
17 )
```

xic-plot

Generates a XIC plot.

Parameters

```
xic-plot(
  width: length relative,
  height: length relative,
  rt-range: auto array,
  max-intensity: none float,
  title: content,
  ..xic-item: dictionary
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length` ; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto` .

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length` ; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto` .

Default: `10cm`

rt-range `auto` or `array`

Data limits along the x-axis (retention time in seconds). Expects auto or a tuple (min, max) where min and max may individually be auto

Default: `auto`

max-intensity `none` or `float`

Maximum intensity to display. **Optional.**

Example: `30000`

Default: `none`

title `content`

Graph title

Default: `none`

..xic-item dictionary

dictionary containing XIC data

```
1  /// XIC dictionary structure typc
2  #let xic_item = (
3    /// Array of retention times values
4    "x": (),
5    /// Array of intensity values
6    "y": (),
7    /// Title for this XIC
8    "title": "title for this XIC"
9    /// Detected peak retention time start. *Optional*.
10   "peak-begin": 3500,
11   /// Detected peak retention time end. *Optional*.
12   "peak-end": 3500,
13 )
```

Retention time alignment

The alignment of retention times between MS runs done with MassChroQ3 are represented as follow in the JSON output

MS run alignment data

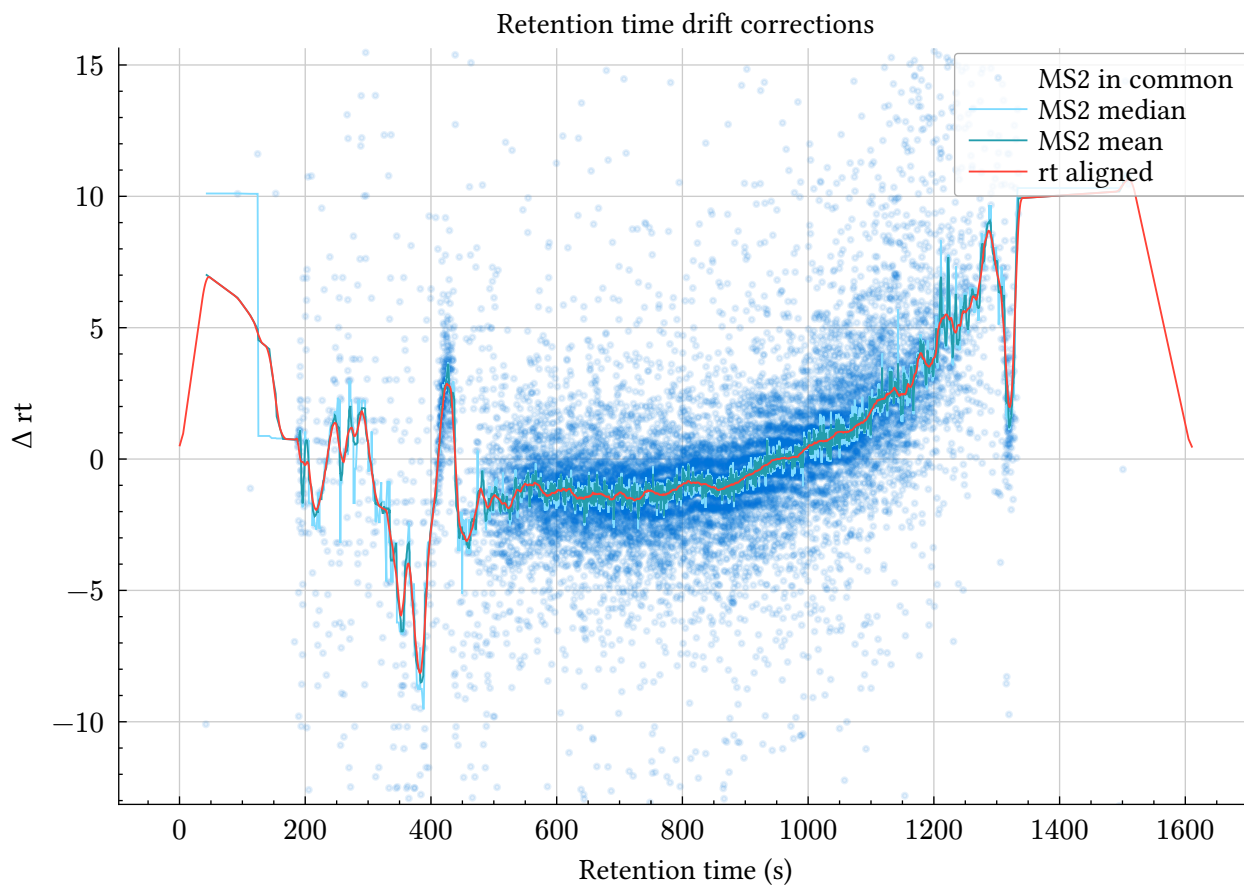
```
{
  "corrections": {
    "msrun1": {
      "aligned": [],
      "ms2_delta_rt": {
        "x": [],
        "y": []
      },
      "ms2_mean": [],
      "ms2_median": [],
      "original": []
    },
    "msrun2": {
      "aligned": [],
      "ms2_delta_rt": {
        "x": [],
        "y": []
      },
      "ms2_mean": [],
      "ms2_median": [],
      "original": []
    },
    "msrun3": {
      "aligned": [],
      "ms2_delta_rt": {
        "x": [],
        "y": []
      },
      "ms2_mean": [],
      "ms2_median": [],
      "original": []
    },
    "msrun_ref": "msrun4"
  }
}
```

Example :

```
#import "@preview/teproteograph:0.2.2": *

#let rt_align = json("../examples/data/one_alignment.json")
#rtalign-plot(title: "Retention time drift corrections", ylim: auto,
  ms2-delta-rt: rt_align.ms2_delta_rt,
  aligned: rt_align.aligned,
  ms2-mean: rt_align.ms2_mean,
  ms2-median: rt_align.ms2_median,
  original: rt_align.original)
```

Result :



Retention time alignment full documentation

- [rtalign-plot\(\)](#)

rtalign-plot

retention time delta plot between MS runs plot.

Parameters

```
rtalign-plot(  
  width: length relative,  
  height: length relative,  
  xlim: auto array,  
  ylim: auto array,  
  title,  
  original,  
  aligned,  
  ms2-delta-rt,  
  ms2-mean,  
  ms2-median  
) -> content
```

width `length` or `relative`

The width of the diagram. This can be

- A `length` ; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with `width: auto` .

Default: `15cm`

height `length` or `relative`

The height of the diagram. This can be

- A `length` ; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A `ratio` or `relative` where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with `height: auto` .

Default: `10cm`

xlim `auto` or `array`

Data limits along the x-axis (retention time in seconds). Expects `auto` or a tuple (min, max) where min and max may individually be `auto`

Default: `auto`

ylim `auto` or `array`

Data limits along the y-axis (retention time delta in seconds). Expects `auto` or a tuple (min, max) where min and max may individually be `auto`

Default: `auto`

MS2 spectra

Simple MS2 spectra

The spectra is a dictionary containing 2 arrays of floats : mz and intensity

spectra dictionary

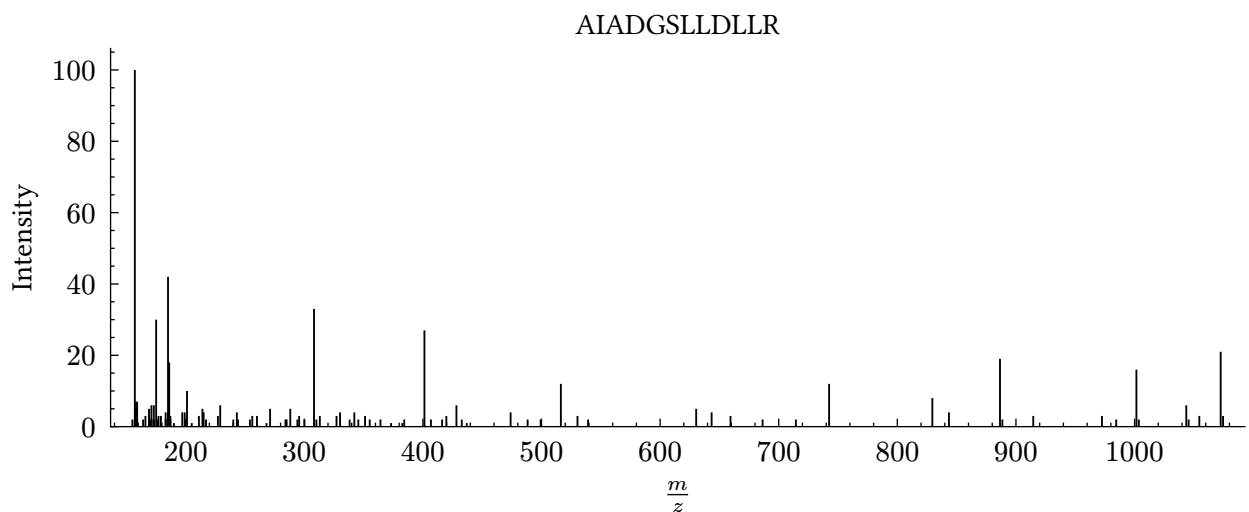
```
{
  "intensity": [
    2,
    2,
    100,
    5,
    7,
    2
  ],
  "mz": [
    155.081,
    157.097,
    157.133,
    158.092,
    158.137,
    159.076
  ]
}
```

Example :

```
#import "@preview/tegraph:0.2.2": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(height: 5cm, title: "AIADGSLLDLLR", spectra: complete_psm.spectra)
```

Result :



MS2 spectra with ion annotations

This example shows an MS2 spectra with ion series annotation

The ion annotations are described in a dictionary :

ion-series dictionary

```
{
  "a": [
    {
      "charge": 1,
      "intensity": 100,
      "mz": 157.133,
      "mzth": 157.13353961189497,
      "size": 2
    }
  ],
  "b": [
    {
      "charge": 1,
      "intensity": 42,
      "mz": 185.128,
      "mzth": 185.12845423145498,
      "size": 2
    },
    {
      "charge": 1,
      "intensity": 3,
      "mz": 256.165,
      "mzth": 256.16556801702,
      "size": 3
    }
  ]
}
```

The first key is the ion series name as follow :

y, b, yp, ystar, y*, yO, x, bstar, b*, bO, a, astar, aO, c, z, bp

and any ion can be described by :

- [ion-description\(\)](#)

ion-description

Ion description

Parameters

```
ion-description(
  "charge": integer,
  "intensity": float,
  "mz": float,
  "mzth": float,
  "size": integer
)
```

"charge" integer

Ion charge

Default: 1

"intensity" float

Ion intensity

Default: none

"mz" float

Ion experimental mass on charge ratio

Default: none

"mzth" float

Ion theoretical mass on charge ratio. **Optional.**

Default: none

"size" integer

Ion size : number of residues for this fragment

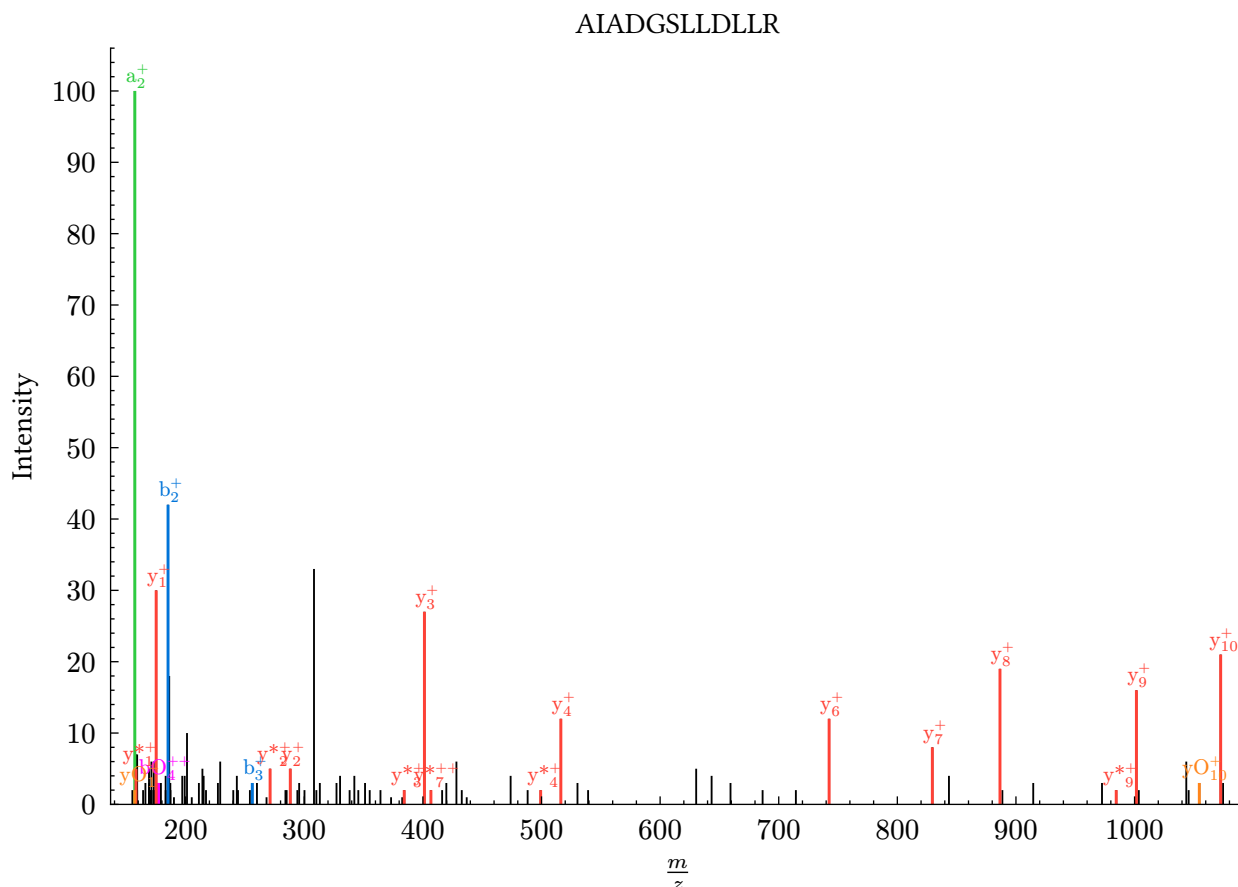
Default: 1

Example :

```
#import "@preview/tegraph:0.2.2": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(title: "AIADGSLLDLLR", spectra: complete_psm.spectra, ion-series:
complete_psm.ion-series)
```

Result :



MS2 spectra zoom (mz_range and intensity)

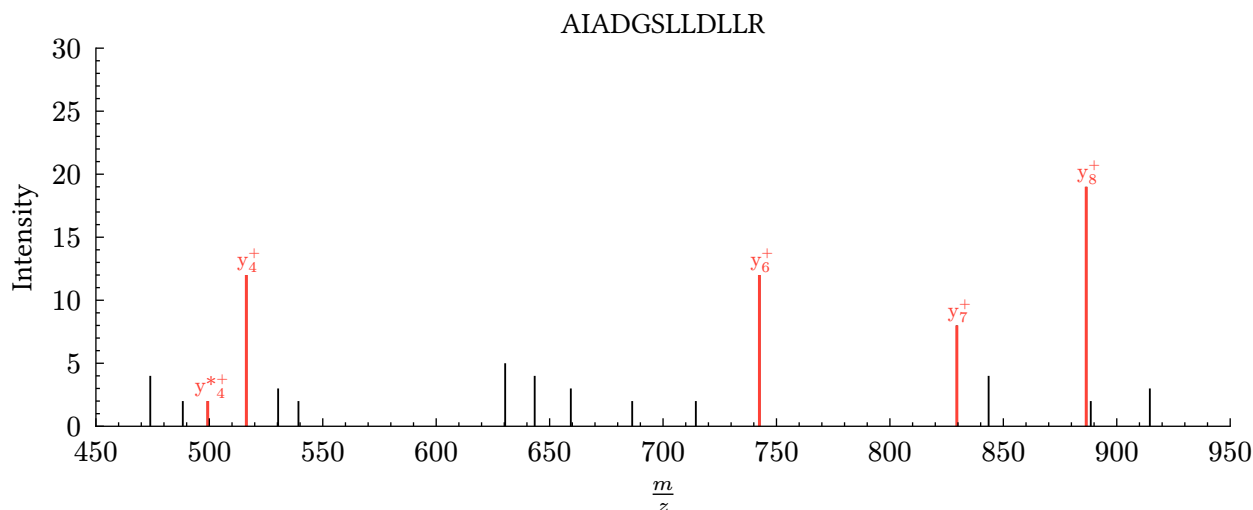
This example shows an MS2 spectra zoomed on a specific m/z range and maximum intensity.

Example :

```
#import "@preview/tegraph:0.2.2": *

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(width: 15cm, height: 5cm, title: "AIADGSLLDLLR", spectra:
complete_psm.spectra, ion-series: complete_psm.ion-series, mz-range: (450, 950), max-
intensity: 30)
```

Result :



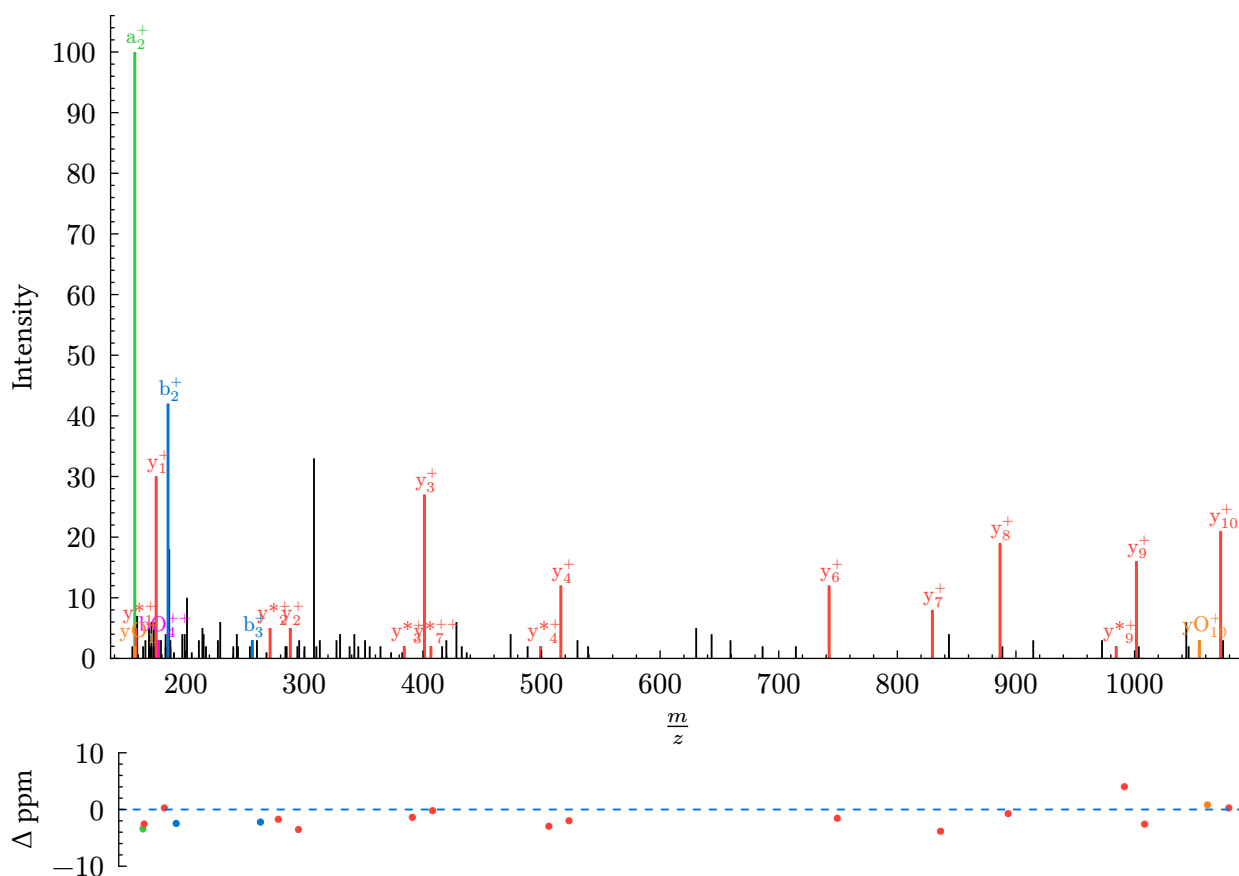
MS2 spectra with ion annotations and MS2 fragments mass delta

This example shows an MS2 spectra with ion series annotation

Example :

```
#import "@preview/proteograph:0.2.2": *  
  
#let complete_psm = json("../examples/data/complete_psm.json")  
#ms2spectra-plot(width: 15cm, height: 10cm, spectra: complete_psm.spectra, ion-series:  
complete_psm.ion-series, delta-fragments: true)
```

Result :



MS2 spectra with ion annotations and highlighted mass delta between peaks

The space between two peaks can be highlighted and manually annotated using an array defined as follow

Highlight space between peaks array

```
[
{
  "ion": "y",
  "mz": [401.287,516.313],
  "level": 1,
  "label": "D"
},
{
  "ion": "y",
  "mz": [742.481,829.511],
  "level": 1,
  "label": "S"
},
{
  "ion": "y",
  "mz": [829.511,886.535],
  "level": 1,
  "label": "G"
},
{
  "ion": "y",
  "mz": [886.535,1001.56],
  "level": 1,
  "label": "D"
}
]
```

Example :

```
#import "@preview/teproteograph:0.2.2": *
```

```
#let delta_arr = (
(
  "ion": "y",
  "mz": (401.287,516.313),
  "level": 1,
  "label": "D"
),
(
  "ion": "y",
  "mz": (742.481,829.511),
  "level": 1,
  "label": "S"
),
(
  "ion": "y",
  "mz": (829.511,886.535),
```

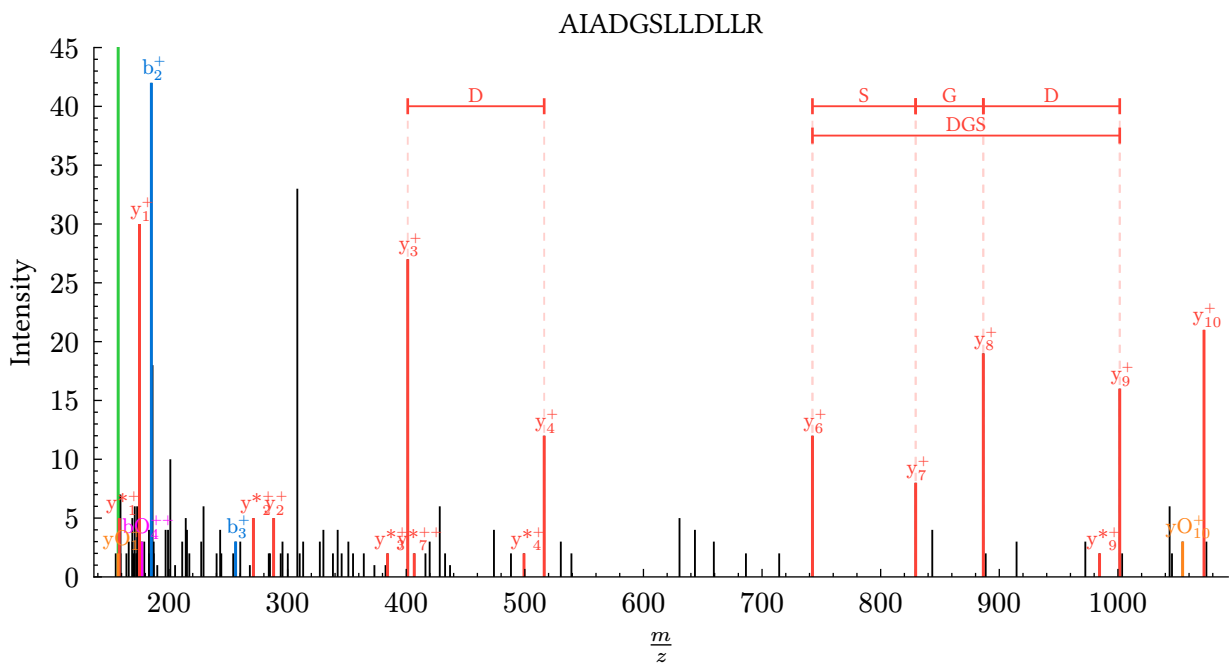
```

    "level": 1,
    "label": "G"
  ),
  (
    "ion": "y",
    "mz": (886.535, 1001.56),
    "level": 1,
    "label": "D"
  ),
  (
    "ion": "y",
    "mz": (742.481, 1001.56),
    "level": 2,
    "label": "DGS"
  )
)

#let complete_psm = json("../examples/data/complete_psm.json")
#ms2spectra-plot(width: 15cm, height: 7cm, max-intensity: 45, title: "AIADGSLLDLLR",
spectra: complete_psm.spectra, ion-series: complete_psm.ion-series, delta: delta_arr)

```

Result :



MS2 spectra full documentation

- [ms2spectra-plot\(\)](#)

ms2spectra-plot

Generates an annotated MS2 spectra plot.

Parameters

```
ms2spectra-plot(  
  width: length relative,  
  height: length relative,  
  title: str content,  
  mz-range: none array,  
  max-intensity: none float,  
  spectra: none dictionary,  
  ion-series,  
  delta,  
  delta-fragments: bool  
) -> content
```

width length or relative

The width of the diagram. This can be

- A length ; in this case, it defines just the width of the data area, excluding axes, labels, title etc.
- A ratio or relative where the ratio part is relative to the width of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded width, e.g., a page with width: auto .

Default: 15cm

height length or relative

The height of the diagram. This can be

- A length ; in this case, it defines just the height of the data area, excluding axes, labels, title etc.
- A ratio or relative where the ratio part is relative to the height of the parent that the diagram is placed in. This is not allowed if the parent has an unbounded height, e.g., a page with height: auto .

Default: 10cm

title str or content

Shows the plot title. **Optional.**

Default: none

mz-range none or array

m/z range to display. **Optional.**

Example: (450, 950)

Default: none

max-intensity `none` or `float`

maximum intensity to display. **Optional.**

Example: `30000`

Default: `none`

spectra `none` or `dictionary`

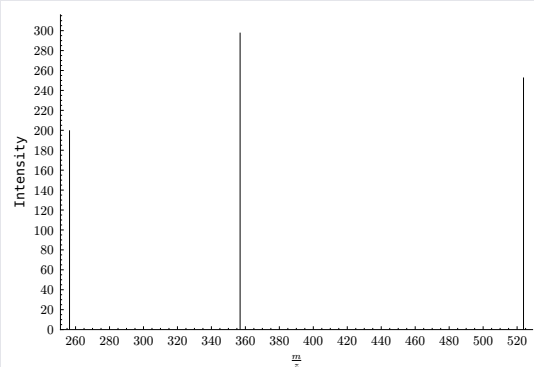
Mass spectra values. **Optional.**

a dictionary with the keys `mz` (array of mass to charge ratios) and `intensity` (array of intensities) with the same length for each array

(e.g., `spectra: (mz: (256.45, 356.89, 523.78), intensity: (200, 298, 253))`)

```
1 #import "@preview/
  proteograph:0.2.2": *
2 #set text(size: 12pt)
  #ms2spectra-plot( spectra: (mz:
3 (256.45, 356.89, 523.78), intensity:
  (200, 298, 253)))
```

typ



Default: `none`

delta-fragments `bool`

Whether to clip the matched ion mass delta to the plot. **Optional.**

Default: `false`

Protein sequence

Example :

```
#import "@preview/teograp:0.2.2": *
```

```
#protein-sequence(line-length: 60,
```

```
"MASTKAPGPGEKHHSIDAQLRQLVPGKVSDDKLEIYDALLVDRFLNQLDHLGPSLREFVQECYEVSADYEGKGDITKLGLGAKLTGLAPADAILV
```

Result :

```
0  MASTKAPGPGEKHHSIDAQLRQLVPGKVSDDKLEIYDALLVDRFLNQLDHLGPSLREF
60  VQECYEVSADYEGKGDITKLGLGAKLTGLAPADAILVASSILHMLNLANLAEVQIAHR
120  RRNSKLKKGFADEGSATTESDIEETLKRVLSEVGKSPPEVFEALKNQTVDLVFTAHTQ
180  SARRSLLQKNARIRNCLTQLNAKDITDDDKQELDEALQREIQAAFRTDEIRRAQPTPQDE
240  MRYGMSYIHETVWKGVPKFLRRVDTALKNIGINERLPYNVSLIRFSSWMGGDRDGNPRVT
300  PEVTRDVCLLARMMAANLYIDQIEELMFELSMWRCNDELRVRAEELHSSSGSKVTYYYIE
360  FWKQIPPNEPYRVILGHVRDKLYNTRERARHLLASGVSEISAESSFTSIEEFLEPLELCY
420  KSLCDCGDKAIADGSLLDLLRQVFTFGLSLVKLDIRQESERHTDVIDAITTHLGIGSYRE
480  WPEDKRQEWLLSELRGKRPLLPDLPQTDEIADVIGAFHVLAEPPDSFGPYIISMATAP
540  SDVLAVELLQRECGVRQPLPVVPLFERLADLQSAPASVERLFSVDWYMDRIKQKQVMVG
600  YSDSGKDAGRLSAWQLYRAQEEMAQVAKRYGVKLTLFHGRGGTVGRGGPHTLAISQP
660  PDTINGSIRVTVQGEVIEFCFGEEHLCFQTLQRFTAATLEHGMHPPVSPKPEWRKLMDM
720  AVVATEEYRSVVVKEARFVEYFRSATPETEYGRMNIGSRPAKRRPGGGITTLRAIPWIFS
780  WTQTRFHLPVWLGVAFAFKFAIDKDVRNFQVLKEMYNEWPFVRVTLDLLEMVFAKGDPGI
840  AGLYDELLVAEELKPFQKLRDKYVETQQLLLQIAGHKDILEGDPFLKQGLVLRNPYITT
900  LNVFQAYTLKRIRDPNFKVTPQPPLSKEFADENKPAGLVKLNPASEYPPGLEDTLILTMK
960  GIAAGMQNTGA
```

Protein sequence documentation

- [protein-sequence\(\)](#)

protein-sequence

Display a protein sequence in a block with amino acid numbering

Parameters

```
protein-sequence(  
  line-length,  
  protein  
) -> content
```

line-length

The maximum length of amino acid to display in a line. **Optional**.

Default: 50

protein

Amino acid sequence (one letter code) of the protein to display

Protein diagram

This figure is based on the [fletcher](#) package features.

Example :

```
#import "@preview/teotopology:0.2.2": *

#let tint(c) = (stroke: c, fill: rgb(..c.components().slice(0,3), 5%), inset: 1pt)

#protein-diag(line-length: 60,
"MASTKAPGPGGEKHHSIDAQLRQLVPGKVSDDKLEIYDALLVDRFLNILQDLHGPSLREFVQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAILV
(pos: (1,5), style: tint(green)),
(pos: (12,17), style: tint(teal)),
(pos: (75,177), style: tint(red)),
)
```

Result :

```
MASTKAPGPGGEKHHSIDAQLRQLVPGKVSDDKLEIYDALLVDRFLNILQDLHGPSLREF
VQECYEVSADYEGKGDTTKLGELGAKLTGLAPADAILVASSILHMLNLANLAEEVQIAHR
RRNSKLLKGGFADEGSATTESDIEETLKRLVSEVGKSPPEVFEALKNOTVDLVFTAHPTQ
SARRSLLQKNARIRNCLTQLNAKDITDDDKQELDEALQREIQAARTDEIRRAQPTPQDE
MRYGMSYIHETVWKGVPKFLRRVDTALKNIGINERLPYNVSLIRFSSWMGGDRDGNPRVT
PEVTRDVCLLARMMAANLYIDQIEELMFELSMWRCNDELVRRAEELHSSSGSKVTKYIE
FWKQIPPNEPYRVILGHVRDKLYNTRERARHLLASGVSEISAESSFTSIEEFLEPLELCY
KSLCDCGDKAIADGSLDLLRQVFTFGLSLVKLDIRQESERHTDVIDAITTHLGIGSYRE
WPEDKRQEWLLSELRGKRPLLPDLPQTDEIADVIGAFHVLAEPLPDSFGPYIISMATAP
SDVLAVELLQRECGVRQPLPVVPLFERLADLQSAPASVERLFSVDWYMDRIKKGQVMVG
YSDSGKDAGRLSAAWQLYRAQEEMAQVAKRYGVKLTLFHGRGGTVGRGGPPTHAILSQP
PDTINGSIRVTVQGEVIEFCFGEEHLCFQTLQRFTAATLEHGMHPPVSPKPEWRKLMDEM
AVVATEEYRSVVVKEARFVEYFRSATPETEYGRMNIGSRPAKRRPGGGITTLLRAIPWIFS
WTQTRFHLPVWLGVGAFAFKFAIDKDVRNFQVLKEMYNEWPFVRVTLDLLEMVFAKGDPI
AGLYDELLVAEELKPFQKLRDKYVETQQLLLQIAGHKDILEGDPFLKQGLVLRNPYITT
LNVFQAYTLKRIRDPNFKVTPQPPLSKEFADENKPAGLVKLNPASEYPPGLEDTLILTMK
GIAAGMQNTGA
```

Protein diagram documentation

- [protein-diag\(\)](#)

protein-diag

Display a protein sequence in a block with amino acid numbering

Parameters

```
protein-diag(
  line-length: int,
  boxes: array,
  protein: str,
  ..box: dictionary
) -> content
```

line-length `int`

The maximum length of amino acid to display in a line. **Optional**.

Default: `50`

boxes `array`

Array of boxes to draw in the protein sequence **Optional**.

This can be used to visualize the position of one or more peptides

example for 3 peptides :

```
((pos: (1,5), style: tint(green)),(pos: (12,17), style: tint(teal)),(pos: (75,177), style: tint(red)))
```

Default: `()`

protein `str`

Amino acid sequence (one letter code) of the protein to display

..box `dictionary`

box item to draw in the protein sequence **Optional**.

This can be used to visualize the position of one or more peptides

example :

```
(pos: (1,5), style: tint(green))
```

```
1 #import "@preview/
2 #proteograph:0.2.2": *
3 "MASTKAPGPGGEKHHSIDAQLRQLV",
4 (pos: (1,5), style: (stroke: red,
5 inset: 1pt)),
6 (pos: (12,17), style: (stroke: teal,
7 inset: 3pt))
8 )
```

MASTKAPGPGGEKHHSIDAQLRQLV

Isotope pattern plot

Example :

```
#import "@preview/proteograph:0.2.2": *
```

```
#isotope-pattern-plot(
  mz: (792.89009986061, 793.3914387289032, 793.8927444414951),
  intensity: (22000, 18000, 8000),
  th-ratio: ( 0.4182709626684006, 0.3428843140088085, 0.157722041765769),
  color: orange,
)
```

Result :

